Automatic Soap Dispenser

Jacob Yehuda, Faith Godden, Joey Arens, Brett Womack
Section A6, Group 1
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Executive Summary:

Our project goal was to build an efficient automatic soap dispenser that dispenses enough soap to wash your hands without ever touching the device. The target audience is employees and students working in EPIC who may have oil or debris on their hands, who don't want to dirty the soap bottle by pumping it. The automatic dispenser is also extremely useful for people with limited fine motor skills. To achieve our goal, we utilised a linear actuator for mechanical force and a laser break sensor for sensing. The sensor is used to detect the presence of the hand, activating the linear actuator, which will push down on the pump and dispense soap. In order to house the linear actuator, we had to mount the motor within a 3D printed component, and the housing had to account for the normal force being applied. We tested reliability and found that our product both sensed the hand present and dispensed enough soap 87% of the time. As testing went on, the pump began to become stiffer, causing an increase in failed tests. As for detection, the sensor detected the user every time, almost instantaneously. Multiple ways of interacting with the sensors were measured. When one's hand is cupped and perpendicular to the arms of the device, there is a distance of approximately 2cm, and when the user's hand is parallel there is approximately 5cm of clearance. There will of course be variability, depending on the size of the user's hand. Our test results indicate that our device detects the user, dispenses soap, and is completely touch-free, meaning we have almost met all of our goals.

Introduction

Our client desired some form of an automatic dispenser of either water, soap, or toothpaste. We determined our problem statement to be that we would design an automatic soap dispenser for EPIC that releases soap with no contact points, to promote better hygiene and convenience. Our key objectives included releasing soap, detecting the user, being non-contact, reliable, easy to clean, and lastly, affordable (**Figure 1**). We utilized a PCC chart in order to determine their priority, so we knew where we needed to focus our time and funds (**Figure 3**). To evaluate our performance in meeting these objectives, we also determined quantifiable metrics. For soap release, we tested that there was a measurable amount of soap dispensed from the container. User detection was timed and needed to be under 2 seconds. To maintain our no-contact design, we specified a minimum distance of 1 cm between the sensor and the user's hand. Reliability was set at a 96% success rate over 30 tests to ensure consistent performance. For ease of cleaning, we ensured it would take less than 1 minute to wipe down the device. Finally, affordability was defined as staying within the $200 budget.

Design Elements

Our design incorporates many elements to improve its function, hygiene, and accessibility. The device is powered by a 12V AC/DC plug (3A) as the linear actuator requires 12V to function. A plug was chosen instead of a rechargeable battery, as we wanted to keep the user interface with the product to a minimum (**Figure 4**). The device will most likely be stationed in one location; therefore, a plug seemed reasonable to utilize. The plug needed to be

able to supply 3A as the linear actuator draws around 2A when moving a considerable amount of load; other standard wattages would not have been sufficient to support the actuator (**Figure 11**).

As for the choice of the Linear Actuator, it was chosen as it was compact and had a linear displacement which was perfect for our products needs. The Linear Actuator was mounted on the inside of the housing with a 3D printed component, to hold it in place (**Figure 9**). The extending arm of the actuator was connected to another 3D printed component that was fit to the top of the GOJO soap pump, as to have a seamless connection between the two (**Figure 10)**.

In order to swap the polarity of the Linear Actuator, an H-Bridge was necessary to include in our circuit. The H-Bridge connects to our Arduino Uno and reverses the polarity when programmed to retract the actuator. The H-Bridge also has two channels of power, 12V which is coming from the plug, and 5V which is powering the Arduino Uno and the sensing component (**Figure 6**). As for deciding the sensing component, multiple options were considered (**Figure 4**). Infrared sensors have been known to struggle with sensing people of color, as we wanted to have an inclusive design, we decided against this. The Laser Break-Beam sensor was perfect, as any object, regardless of shape, size, or color, can be detected. The Lasers are mounted as an extension of the main body, sitting around where a hand would receive the soap.

Our Housing was crucial to our design; The main objectives with our housing were: manufacture by laser cutting for ease of scaled production, and support the force supplied by our linear actuator to our soap pump. The combination of these two goals was a challenge because the use of traditional finger joints to bind laser-cut pieces together would break when force was applied parallel to the finger joint. So we had to develop a T-finger joint design that would allow our housing to withstand the force exerted by our linear actuator (**Figure 8**). This design was much more structurally sound but came with a lot of design challenges. We ended up creating three prototypes before we finally settled on a final housing. The added clearances at the top and bottom of the finger joints provided an extra level of testing and precision to ensure all the parts fit together.

As for the material used for the housing, acrylic was determined to be the best choice. The material needed to be strong enough to withstand the normal force from the linear actuator and waterproof. Acrylic also improves the ease with which a user can wipe down the device, which was one of our metrics (**Figure 13**). Lastly, and most importantly, the Arduino Uno, is the brains of the device. The Arduino receives data from the sensor indicating the presence of the hand, extending the actuator, and retracting it back to its original position (**Figure 7**).

Final Product

The final product uses a sensor to detect a user's hand and automatically actuate the pump mechanism, dispensing soap without coming into contact with the design. The inputs of our product are a 12-volt AD/DC plug, the IR Break Beam Sensor, and a GOJO soap bottle (**Figure 2**). The plug and the soap bottle only need to be installed when the device is first set up. All of our objectives were kept in mind during the design process, incorporating user-friendly features so that the device is easily utilized. Testing across our defined performance metrics demonstrated that the final prototype successfully met almost all of our objectives. The reliability

of the device could be improved, as it currently is only 87% reliable. The total cost of our product was $152.29, keeping us within our allocated budget (**Figure 5**).

Results

Multiple tests were performed to measure our stated objectives. To test the amount of soap was dispensed, we used a small graduated cylinder measuring how much soap was pumped per use. We found that, out of 26 trials, approximately 0.5mL was dispensed, meeting our metric of having more than 0mL dispensed (**Figure 12**). Regarding detection time, we found during circuit development that the sensor response was consistently accurate and nearly instantaneous, too fast to measure manually with precision. From the moment a hand was placed between the sensors until soap dispensed, the entire process took 2.5 seconds (an average over 10 trials), which exceeded our initial expectations for system responsiveness (**Figure 14**). Another objective we wanted to achieve was no contact, which we determined to be at least 1cm of distance from the user to the sensor. We completed four trials with four different users and found a range between 4cm and 6cm of distance between the user's hand and the sensor, certainly meeting our metric (**Figure 15**). As for the time it takes to clean the device, over ten trials of removing the soap bottle and wiping down the eternity of the device, we found a standard distribution centered around 28 seconds, meeting our metric of being under 1 minute (**Figure 13**). Affordability was met as the device's total cost was $152.29, which is under our budget of $200. Lastly, reliability was tested over the course of thirty trials. We determined a successful trial to be that the device worked as intended, with enough soap to wash one's hands. When tested we found a success rate of 86.66%, which is below our described 96% success rate (**Figure 16**). We are confident that with the implementation of a linear actuator with a stronger actuating capacity our success rate would improve.

Lessons Learned

Overall, this project was filled with lessons that were learned. Communication ended up being one of the most important. It became a necessity to communicate deadlines to each other and collaborate on a time line that did not cause a back log of work. Another lesson that was learned was how to use an H-bridge and how to ensure all parts are sufficiently powered. Another major lesson learned was how to laser cut, and create a design that can withstand force. With this lesson came lots of innovation and revision. Finally came writing the code and adjusting it for a real world project, which was also something that had not been explored yet. Assembling everything correctly and accurately taught the team to be patient. Finally being flexible was the key to completing the project smoothly.

References

Actuonix Motion Devices. (2016). *100mm L12 actuator: Miniature linear motion series* [Product datasheet]. RS Components. https://docs.rs-online.com/b02c/0900766b816f733a.pdf

DFRobot. (n.d.). Gravity: I2C Infrared CO2 Sensor (400-5000 ppm) [Product documentation]. Digi-Key. https://mm.digikey.com/Volume0/opasdata/d220001/medias/docus/5344/SEN0503_Web.pdf

Appendix

Figure 1: Objectives and Metrics

Objective	Metric
Release Soap	x>0 soap released from container
Detects Person	x<2 sec detection
No Contact	x>1cm from hand
Reliable	96% realiable for 30 tests
Easy to clean	x<1 minute to wipe down
Affordable	Cost < $200

Figure 2: Glass Box

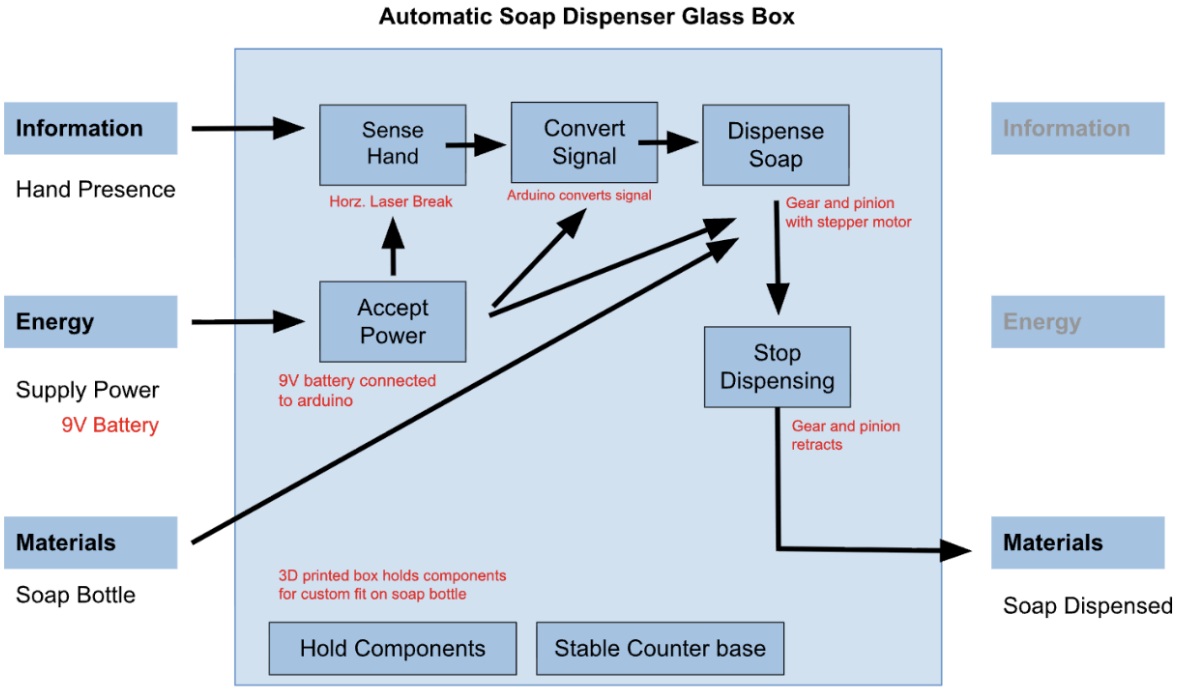


Figure 3: PCC

	Release Soap	No Contact	Reliable	Detects person	Affordable	Easy to clean	Total
Release Soap	-	1	1	1	1	1	5
No Contact	0	-	1	0	1	1	3
Reliable	0	0	-	0	1	1	2
Detects person	0	1	1	-	1	1	4
Affordable	0	0	0	0	-	0	0
Easy to clean	0	0	0	0	1	-	1

Figure 4: Morph Chart

Function	Option 1	Option 2	Option 3
Detect user	Infrared sensor	Ultrasonic Sensor	Break Beam Sensor
Power	Batteries	Wall Plug	Rechargeable Battery
Housing Material	Wood	Metal	Acrylic
Dispense Soap	Pump	Gravity	Pressured Spray

Figure 5: Bill of Materials

Part	Cost
Linear Actuator	$33.99
Arduino Uno	$27.60
Arduino Uno Cable	$7.00
12V AC/DC Adapter	$7.91
24x24in Sheet of 1/8th Black Acrylic	$37.07
IR Break beam sensor	$6.00
Soap	$32.72
Total	$152.29

Figure 6: Electronics Sketch

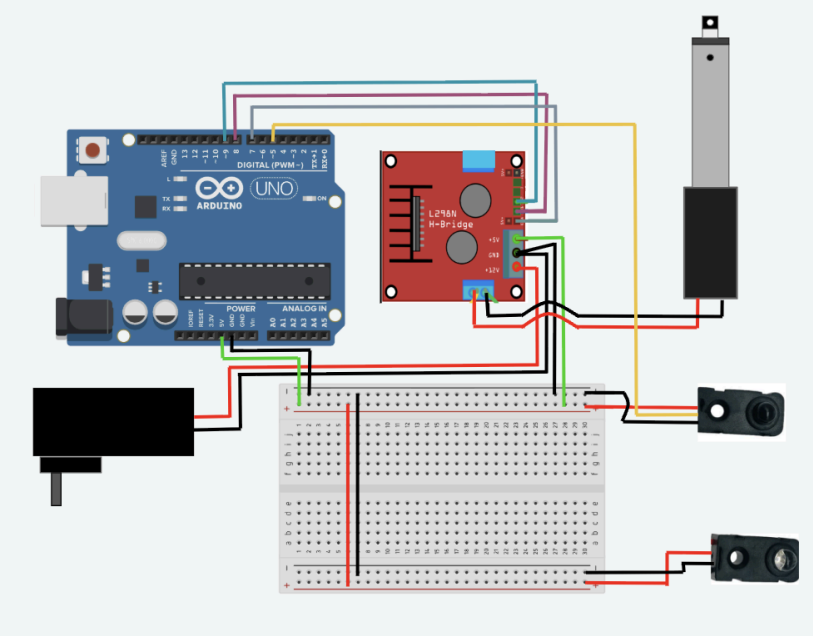


Figure 7: Code Flow Chart

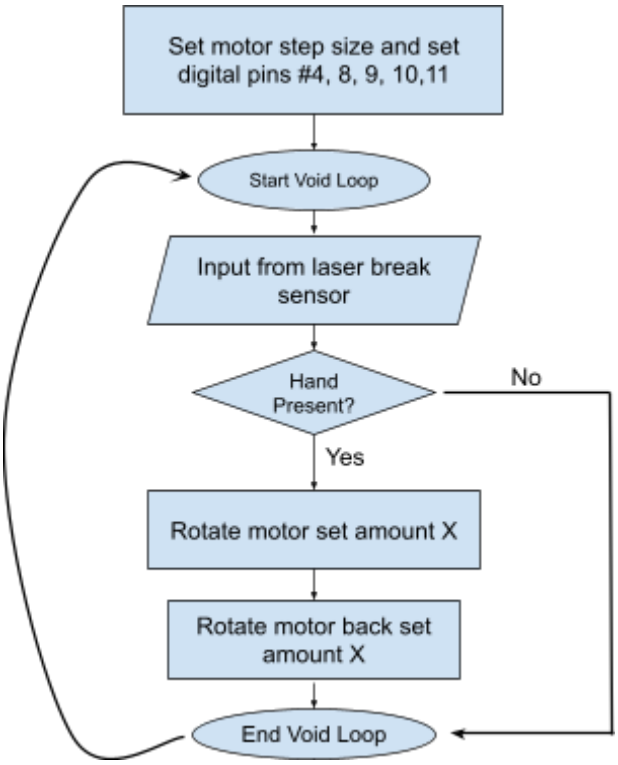


Figure 8: Housing CAD in mm

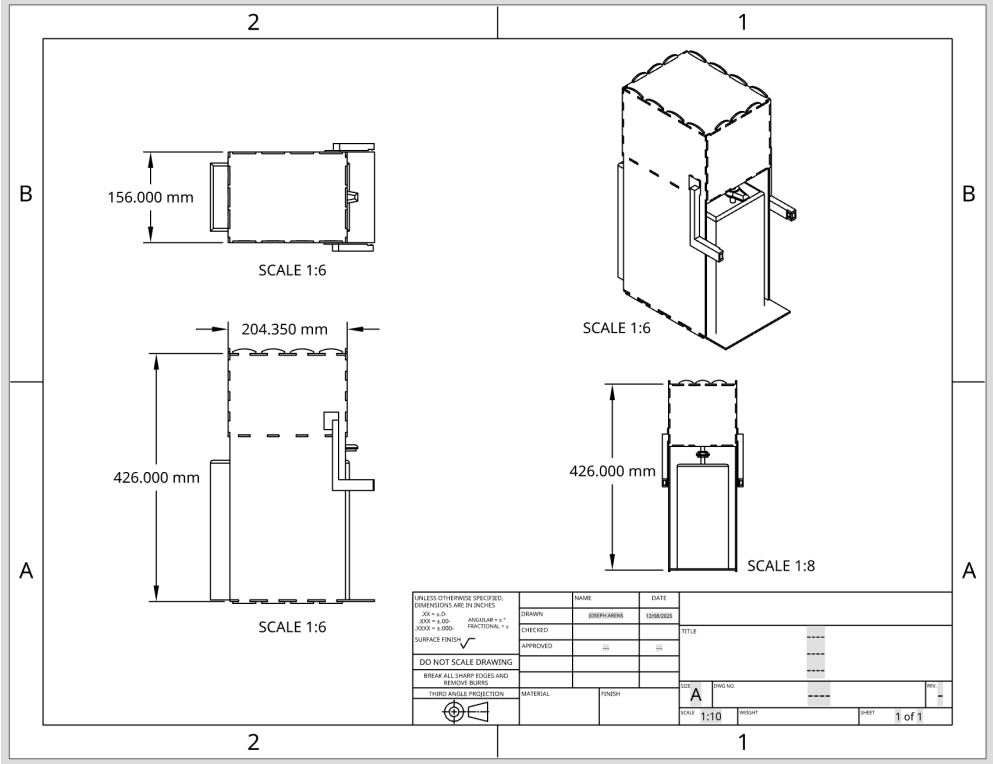


Figure 9: Linear Actuator CAD in mm

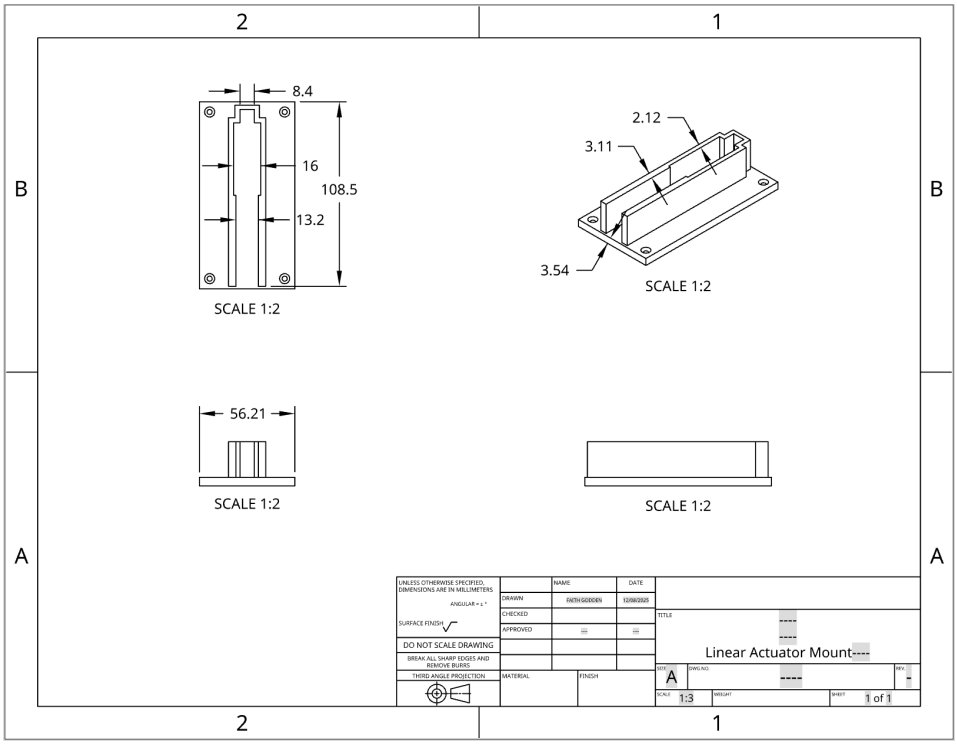


Figure 10: Soap Cap CAD in cm

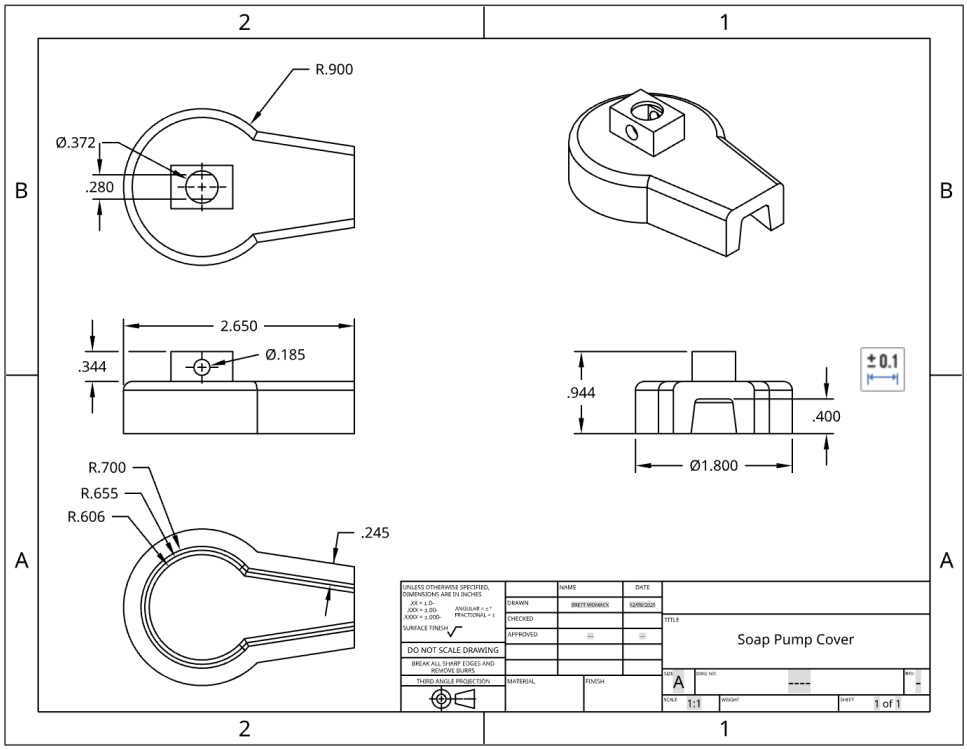


Figure 11: Power Budget Calculations

Component	Voltage (V)	Current (A)	Power (W)
Linear actuator	12.00	2	24
H-Bridge (motor driver)	12.00	0.2	2.4
Arduino	5.00	0.05	0.25
IR break beam sensor	5.00	0.02	0.1
Total Power Drawn			26.75
Power source: 12V AC/DC Plug	12.00	3	36
Remaining Power (Power supplied - Power drawn)			9.25

Figure 12: Amount Dispensed

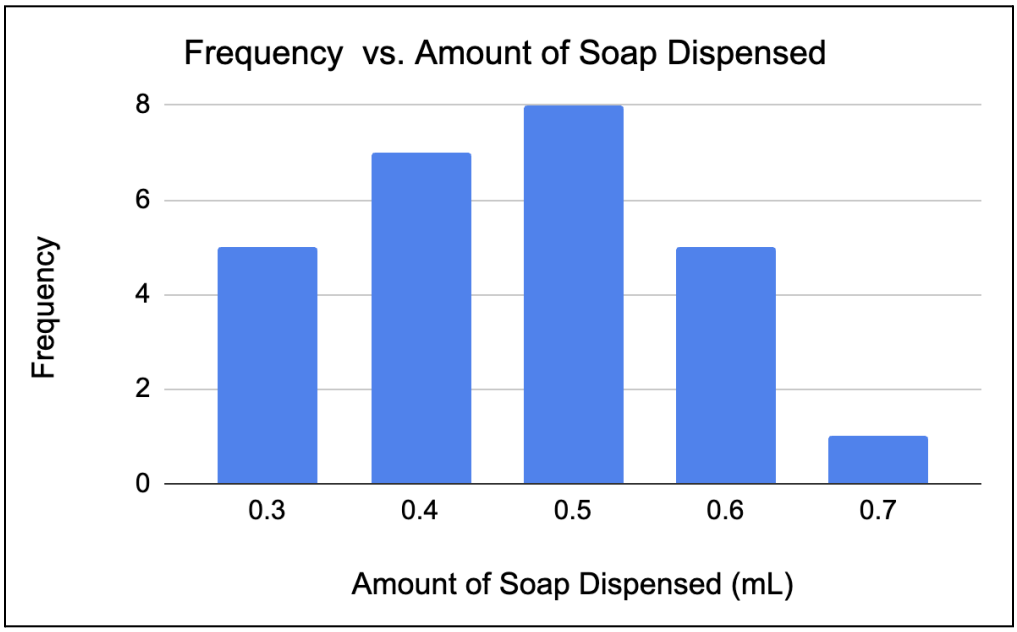


Figure 13: Time to Clean

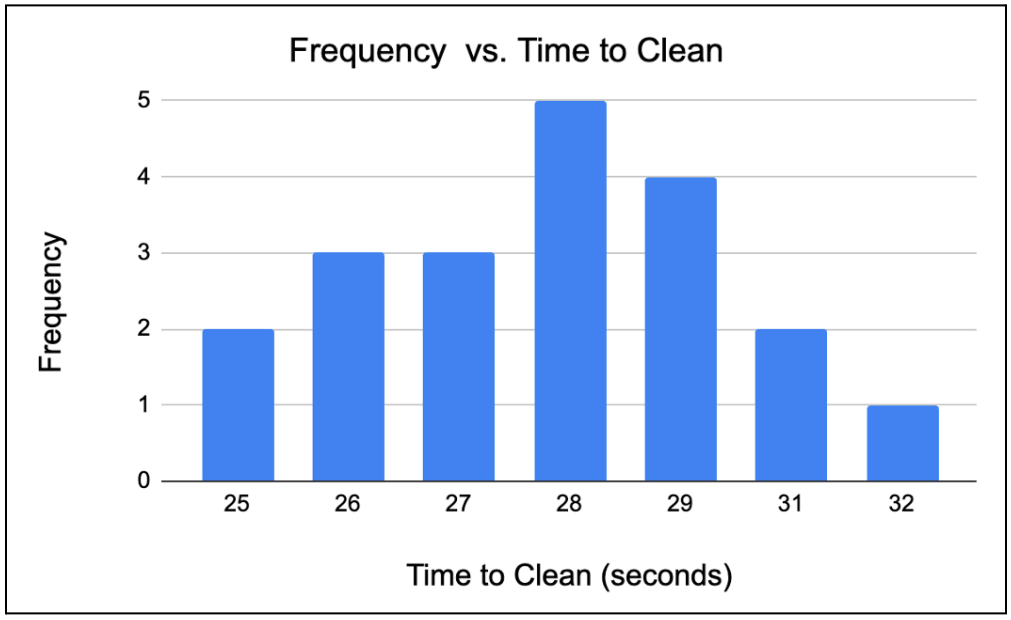


Figure 14: Time to Dispense

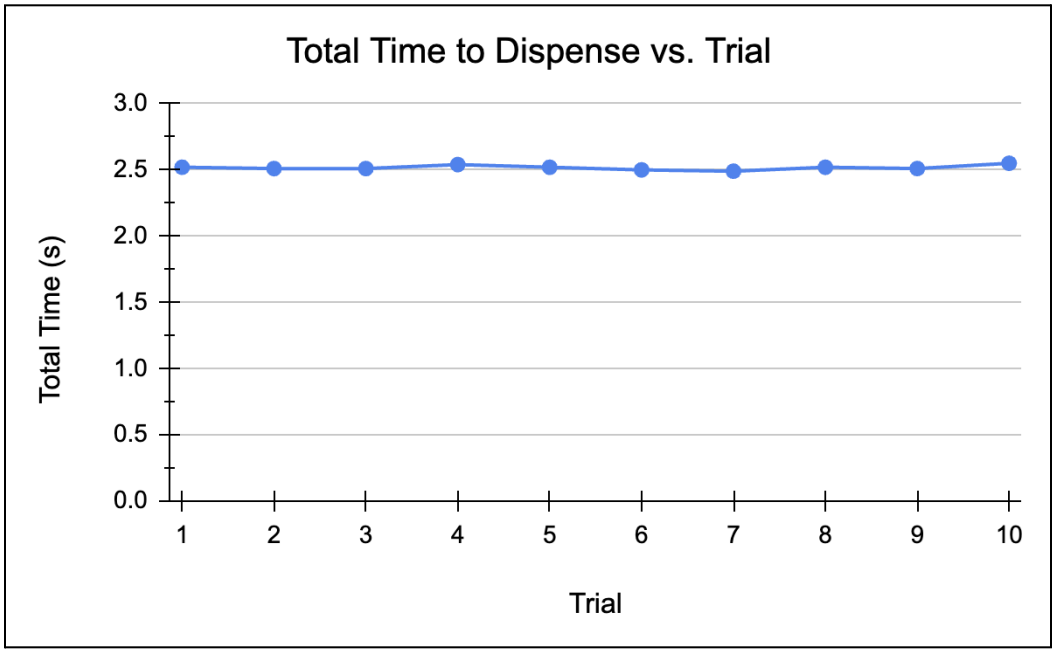


Figure 15: Distance from Hand to Sensor

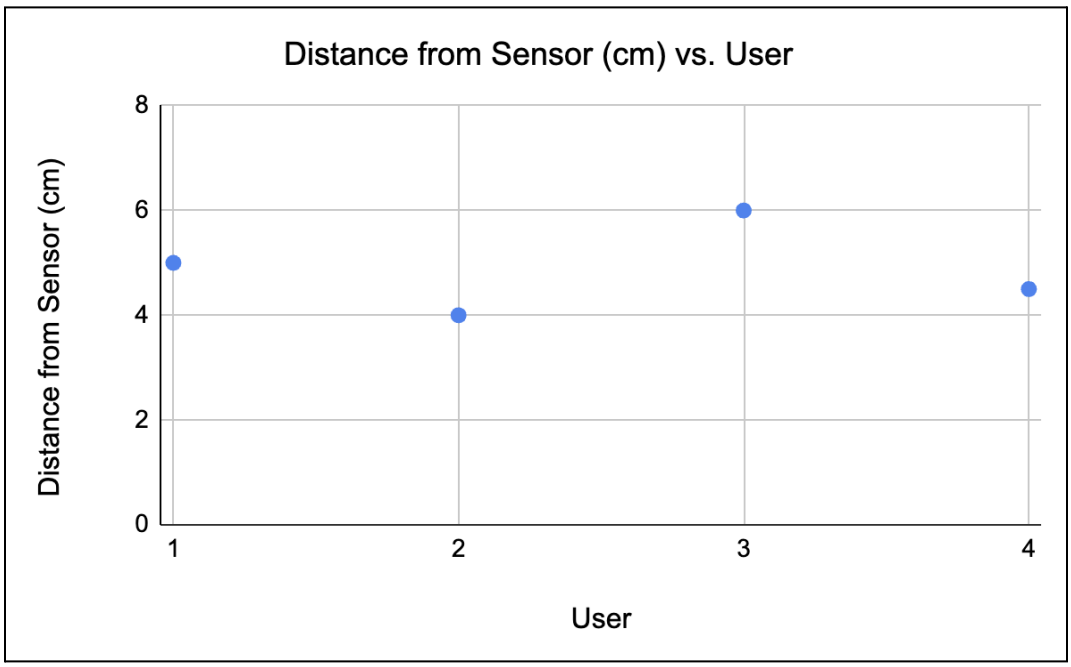


Figure 16: Reliability of Dispenser

